



Proteomics Data Curation Methods

Summary

The following document describes the curation process for the PPMI Proteomics data files listed in Table 1.

Table 1. PPMI Proteomics Datasets

Project ID	Project Description	Analyte	Matrix	# of Samples	File Names	Project Documentation
151	PI: Fiona Elwood , PhD, Novartis Title: Identification of Proteins & Protein Networks in Human CSF That Differentiate Within PD Patients	SomaLogic 4k panel (w/ Merck & Novartis)	CSF	1,158 ¹	Project_151	PPMI_Methods_Project_151_pqtl_Analysis_20210303.pdf
177	PI: Jennifer Van Eyk , PhD, Cedars-Sinai Title: Post-Translational Modification & Protein Quantification of Plasma & CSF	Unbiased DIA-MS (AMP PD)	CSF Plasma	2,283 CSF 949 Plasma	Project_177_csf Project_177_plasma	PPMI_Project177_Proteomics_Methods.pdf
190	PI: Sebastian Virreira Winter PhD OmicEra Title: Targeted & Untargeted MS Based Proteomics of Urine in PD	OmicEra (MS) Urine proteomics	Urine	1,164 ²	Project_190	PPMI_190_Methods_Targeted_Untargeted_MS-based_proteomics_of_urine.pdf
9000	PI: Aparna Vasanthakumar Title: Bridged PPMI Project 196 and 222 Olink® Explore 1536 targeted proteomics assessment of CSF and blood plasma	Olink Explore 1536 (Abbvie and AMP PD DMC)	CSF Plasma	799 CSF 1,031 Plasma	Project_9000_csf Project_9000_plasma	PPMI_Methods_Project_9000_20240216.pdf

¹ Curated dataset for Project 151 includes results for 1,153 CSF samples.
² Curated dataset for Project 190 includes results for 1,162 urine samples.

Curation Process

Data were transformed from the source (LONI) data files into six curated dataset files (one for each platform × matrix).

Source data on LONI				Curated datasets		
Project ID	CSV files	Variables	Observations*	Excel files	Variables	Observations*
151	7	14	5,541,030	1	4,787	1,153
177 (CSF)	1	14	664,353	1	293	2,283
177 (Plasma)	1	14	176,514	1	188	949
190	5	13	4,341,293	1	6,489	1,162
9000 (CSF)	4	13	1,176,115	1	4,391	799
9000 (Plasma)	4	13	1,515,986	1	4,391	1031
* One row per individual test result				* One row per sample		

Note: during this process, the data values were not modified in any way (e.g., no additional QC, normalization, or scaling were performed). Test values reflect those in the source (LONI) data files.



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Each curated dataset shares a common format. Each contains one row per sample, i.e., one row for each unique combination of the PPMI participant ID (PATNO) and visit ID (EVENT_ID). Variable names adhere to a common nomenclature (described in detail below). In each case, a lookup table is provided that matches each variable name to additional data of interest.

Curated nomenclature

Projects used different nomenclature to identify which protein(s) or protein subunit(s) were associated with a given test result. To align, the original labels were converted to a common “gene symbol” format. Each unique analyte was assigned a new label that concatenated the project ID, matrix, and gene symbol. The following table illustrates an example of this process.

Project ID	Matrix	Original nomenclature	Example of conversion to curated nomenclature	
151	CSF	SOMAmer sequence ID	“8288-27_3”	→ “P151_CSF_APOH”
177	CSF	Uniprot ID	“P02749”	→ “P177_CSF_APOH”
177	Plasma	Uniprot ID	“P02749”	→ “P177_Plasma_APOH”
190	Urine	Gene symbol & Uniprot ID	“APOH_P02749”	→ “P190_Urine_APOH”
9000	CSF	Gene symbol	“APOH”	→ “P9000_CSF_APOH”
9000	Plasma	Gene symbol	“APOH”	→ “P9000_Plasma_APOH”

In certain scenarios, the curated nomenclature deviates from this format:

1. If multiple analytes tagged the same protein, then an additional identifier was appended to the variable label.
 - Example: in Project 151, two different SOMAmers (“7915-31_3” and “10627-87_3”) tagged the APLP2 protein. To differentiate, the SOMAmer sequence ID was appended to the end of each variable label, with one labeled “P151_CSF_APLP2_7915_31_3” and the other labeled “P151_CSF_APLP2_10627_87_3.”
2. If one analyte tagged 2 or 3 different proteins, then multiple gene symbols were included in the variable label.
 - Example: in Project 151, SOMAmer sequence ID “2753-2_2” tagged 3 proteins (C1QA, C1QB, C1QC) and was labeled “P151_CSF_C1QA_C1QB_C1QC.”
 - Note: this approach occasionally resulted in a label that exceeded the SAS variable name limit of 32 characters; in such cases, the strategy described below under scenario #3 was applied.
3. If one analyte tagged more than 3 proteins and/or a label with all gene symbols exceeded the SAS variable name limit of 32 characters, then the word “MULTIPLE” combined with a unique identifier was inserted into the variable label (in place of the gene symbols).
 - Example: in Project 151, SOMAmer sequence ID “4179-57_3” tagged 7 proteins (SFN, YWHAB, YWHAE, YWHAG, YWHAH, YWHAQ, YWHAZ) and was labeled “P151_CSF_MULTIPLE_4179_57_3.”

Of note, all lookup tables include “num_genes” and “gene_symbol” variables indicating the number of proteins tagged and the name(s) of the corresponding gene symbol(s), respectively.



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Overlap across proteomics platforms

As part of the curation process, the PPMI Statistics Core created a visit-level data file summarizing the overlap across proteomics platforms. This file, which will be made available on LONI, contains one row for each unique combination of PATNO and EVENT_ID that is associated with results from one or more proteomics datasets. In addition to PATNO and EVENT_ID, it also includes the following variables:

Variable	Description	Code
P151_CSF	Is visit included in project 151 dataset?	0 = No; 1 = Yes
P177_CSF	Is visit included in project 177 (CSF) dataset?	0 = No; 1 = Yes
P177_Plasma	Is visit included in project 177 (plasma) dataset?	0 = No; 1 = Yes
P190_Urine	Is visit included in project 190 dataset?	0 = No; 1 = Yes
P9000_CSF	Is visit included in project 9000 (CSF) dataset?	0 = No; 1 = Yes
P9000_Plasma	Is visit included in project 9000 (plasma) dataset?	0 = No; 1 = Yes

Combining proteomics data with other PPMI variables

To obtain other data of interest (e.g., demographics, SAA results, genetic status), we recommend that users download the “Curated Data Cut” file (created by the PPMI Statistics Core and available on LONI) and merge it by PATNO and EVENT_ID with the curated proteomics datasets described in this documentation.



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The following sections clarify some key project-specific information:

PPMI Project 151: SomaLogic

LONI download date: 12-08-2023

- At the time of download, two versions of source data were available on LONI. The “Batch Corrected” (2022-05-11) version was downloaded from LONI and used for the curated dataset. This included seven CSV files:
 - Project_151_pQTL_in_CSF_1_of_7_Batch_Corrected__08Dec2023.csv
 - Project_151_pQTL_in_CSF_2_of_7_Batch_Corrected__08Dec2023.csv
 - Project_151_pQTL_in_CSF_3_of_7_Batch_Corrected__08Dec2023.csv
 - Project_151_pQTL_in_CSF_4_of_7_Batch_Corrected__08Dec2023.csv
 - Project_151_pQTL_in_CSF_5_of_7_Batch_Corrected__08Dec2023.csv
 - Project_151_pQTL_in_CSF_6_of_7_Batch_Corrected__08Dec2023.csv
 - Project_151_pQTL_in_CSF_7_of_7_Batch_Corrected__08Dec2023.csv
- SOMAmer sequence IDs were matched to gene symbols using an annotation file available on LONI:
 - PPMI_Project_151_pqtl_Analysis_Annotations_20210210.csv
- One curated dataset was generated. This file includes:
 - Results tab
 - 1,153 rows of data (one row per CSF sample) from 1153 participants
 - 4,787 variables
 - PATNO
 - EVENT_ID
 - Test values for 4,785 unique analytes (units = log2 RFU)
 - Lookup tab with additional information pertaining to each analyte
- Key points of clarification
 - Five participants were excluded from the final curated dataset because they were missing an enrollment date (i.e., were not ultimately enrolled into PPMI Clinical). Accordingly, only 1,153/1,158 samples in the source data files are included.
 - Six SOMAmers were identified that corresponded to organism = “Mouse” (rather than organism = “Human”). In such cases, the variable name was modified to include the word “Mouse” (e.g., “P151_CSF_Lrch4_Mouse”) and the lookup tab indicates a value of “Mouse” in the “organism” column.
 - Where necessary to ensure unique variable names, the SOMAmer sequence ID was appended to the end of the variable name.



PPMI Project 177: AMP PD

LONI Download Date: 12-08-2023

- At the time of download, only one version of the source data was available on LONI. This included 1 CSV file:
 - PPMI_Project_177_Untargeted_Proteomics_08Dec2023.csv
- UniProt IDs were mapped to gene symbols using UniProt website (www.uniprot.org/id-mapping). The dataset used to match the UniProt ID to the gene symbol is included in the last tab of the curated project 177 datasets.
- Two curated datasets were created (one for CSF data, one for plasma data). The datasets include:
 - Results tab (CSF file)
 - 2283 rows of data (one row per CSF sample) from 482 participants
 - 293 variables
 - PATNO
 - EVENT_ID
 - Test values for 291 unique analytes (units = Arbitrary Protein Abundance)
 - Results tab (plasma file)
 - 949 rows of data (one row per plasma sample) from 179 participants
 - 188 variables
 - PATNO
 - EVENT_ID
 - Test values for 186 unique analytes (units = Arbitrary Protein Abundance)
 - Lookup tab with additional information pertaining to each analyte
 - Tab with "UniProt Matching File" (used to map UniProt IDs to gene symbols)
- Key points of clarification
 - Where necessary to ensure unique variable names, the UniProt ID was appended to the end of the variable name.



PPMI Project 190: OmicEra

LONI download date: 12-08-2023

- At the time of download, only one version of the source data was available on LONI. This included 5 CSV files:
 - Targeted___untargeted_MS-based_proteomics_of_urine_in_PD_1_of_5_08Dec2023.csv
 - Targeted___untargeted_MS-based_proteomics_of_urine_in_PD_2_of_5_08Dec2023.csv
 - Targeted___untargeted_MS-based_proteomics_of_urine_in_PD_3_of_5_08Dec2023.csv
 - Targeted___untargeted_MS-based_proteomics_of_urine_in_PD_4_of_5_08Dec2023.csv
 - Targeted___untargeted_MS-based_proteomics_of_urine_in_PD_5_of_5_08Dec2023.csv
- Gene symbols were extracted directly from the nomenclature provided by OmicEra.
- One curated dataset was generated. This file includes:
 - 1,162 rows of data (one per urine sample) from 989 participants
 - 6,489 variables
 - PATNO
 - EVENT_ID
 - Test values for 6,487 analytes (units = arbitrary LFQ intensity OR amount [fmol] per 200ng total protein)
- Key points of clarification
 - Consistent with the source (LONI) data files, some analytes have missing values.
 - One sample (PATNO = "3662" at EVENT_ID = "BL") had two sets of test results in the source (LONI) data. These results were excluded from the final curated dataset. Accordingly, only 1,162/1,164 samples in the source data files are included.
 - Where necessary to ensure unique variable names, a 3-digit numeric string (001, 002, etc.) was appended to the end of the variable name.



PPMI Project 9000: Olink

LONI Download Date: 02-23-2024

- At the time of download, only one version of the source data was available on LONI. This included 8 CSV files:
 - PPMI_Project_9000_Plasma_INF_NPX_23Feb2024.csv
 - PPMI_Project_9000_Plasma_NEURO_NPX_23Feb2024.csv
 - PPMI_Project_9000_Plasma_ONC_NPX_23Feb2024.csv
 - PPMI_Project_9000_CSF_Cardio_NPX_23Feb2024.csv
 - PPMI_Project_9000_CSF_INF_NPX_23Feb2024.csv
 - PPMI_Project_9000_CSF_NEU_NPX_23Feb2024.csv
 - PPMI_Project_9000_CSF_ONC_NPX_23Feb2024.csv
 - PPMI_Project_9000_Plasma_Cardio_NPX_23Feb2024.csv
- Gene symbols were extracted directly from nomenclature (i.e., "ASSAY" variable) provided by Olink.
- Two curated datasets were created (one for CSF data, one for plasma data). The datasets include:
 - Results tab
 - CSF data file includes 799 rows of data (one per CSF sample) from 260 participants
 - Plasma data file includes 1031 rows of data (one per plasma sample) from 275 participants
 - Both files include 4391 variables
 - PATNO
 - EVENT_ID
 - Three sets of test results for each of 1,463 analytes
 - Normalized protein expression (NPX) values
 - Fraction of data beneath the limit of detection (MISSINGFREQ) values
 - Binary Olink quality control (QC_WARNING) values
 - Lookup tab with additional information pertaining to each analyte
- Key points of clarification
 - As described above, three sets of test results are provided for each analyte: variables containing NPX values are appended with "_NPX"; those containing MISSINGFREQ values are appended with "_MISS"; and those containing QC_WARNING values are appended with "_QC"

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